High Dynamic Range (HDR) Imaging

Bridging the Gap Between Sensor Physics and Human Perception.



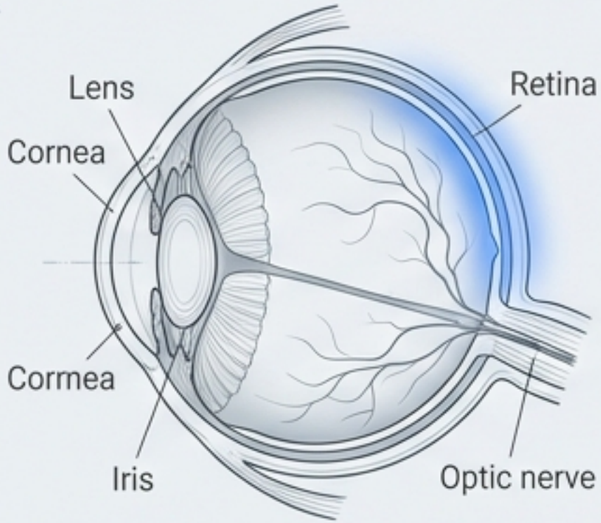
Chapter 12:
Mobile Computational Photography

Mobile sensors face a fundamental disconnect: the real world has infinite contrast, but silicon wells have finite capacity. This presentation explores how computational photography overcomes physical saturation limits to replicate the dynamic range of the human eye.

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The Fundamental Disconnect: Physics vs. Biology

The Biological Eye

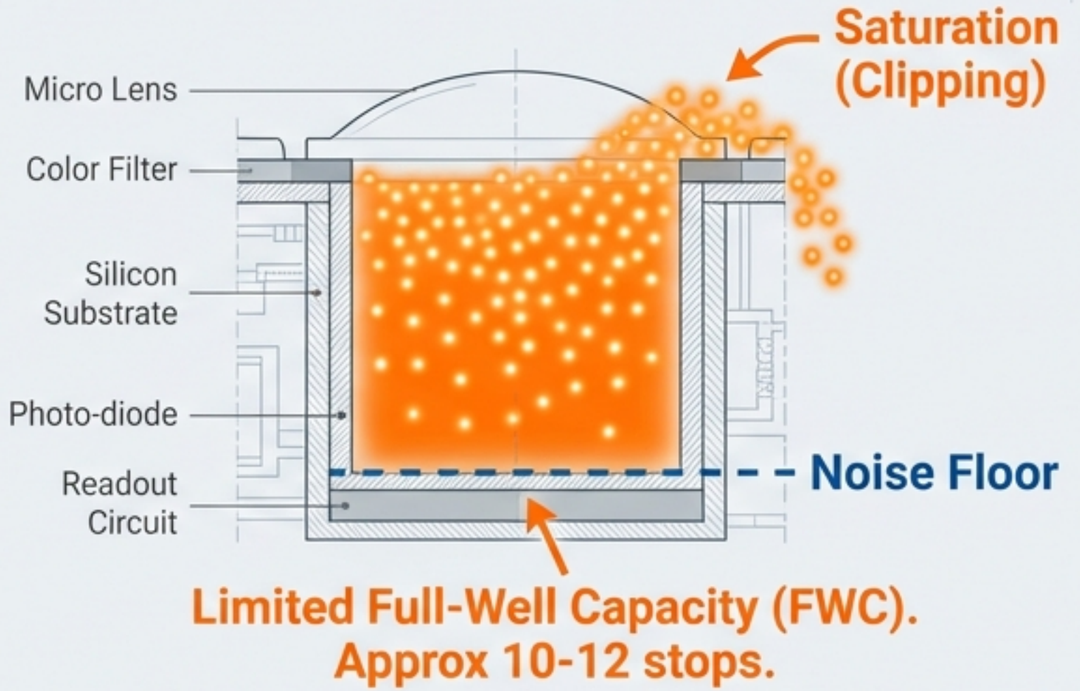




Adaptive Dynamic Range. The eye continuously adjusts to local light conditions, allowing for simultaneous perception of deep shadows and bright highlights.

VS

The Mobile Sensor



Fixed Dynamic Range. The sensor has a linear response to light with a fixed maximum capacity before clipping occurs.

The Gap: The human eye adapts locally to contrast. A sensor is linear and limited. If the bucket is too small, highlights are lost. If the light is too low, the signal is buried in noise. HDR is the computational bridge that artificially expands this bucket.

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The Evolution of Mobile HDR

Early 2010s: Software HDR

- Multi-Exposure Merging
- 3-Frame Capture
- CPU Processing

Key Limitations

- ⚠ Slow Processing,
- ⚠ Ghosting, Limited
- ⚠ Range.

Mid-2010s: Hardware-Assisted

- Enhanced Sensors
- Real-time Preview
- ISP Acceleration

Improvements

- ✓ Fast Readout,
- ✓ Reduced Ghosting.

Present: Computational HDR

- AI-Powered Algorithms
- Semantic Segmentation
- Deep Learning Fusion

Outcome

- ★ Professional-grade images, Zero Shutter
- ★ Lag.

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The Pipeline Step 1: Multi-Frame Compositing

Underexposed
(Short Exposure)

Normal Exposure
Balanced for **Mid-tones**.

Overexposed
(Long Exposure)



Fast shutter speed.
Preserves **Highlights**
(Clouds, Lights).

Slow shutter speed.
Recovers **Shadows**.

Slow shutter speed.
Recovers Shadows.

Zero Shutter Lag (ZSL)

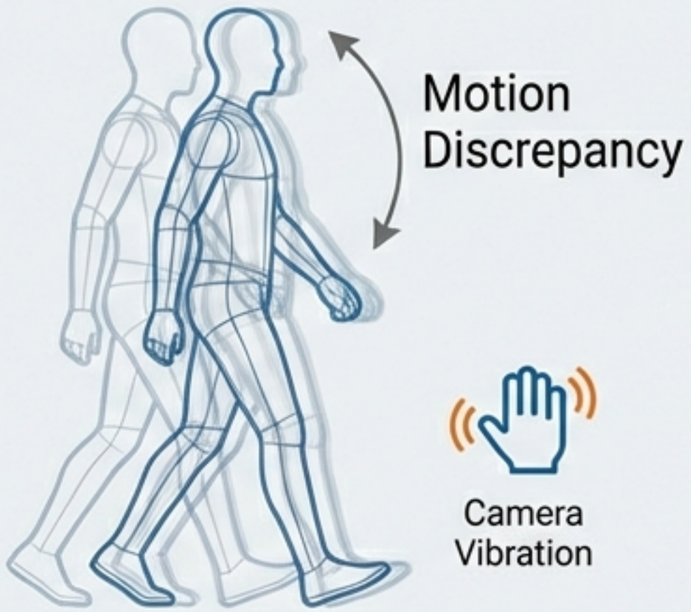
Modern cameras capture these frames continuously into a circular buffer before the button is even pressed.

The 'Burst' provides the raw data foundation that a single exposure cannot.

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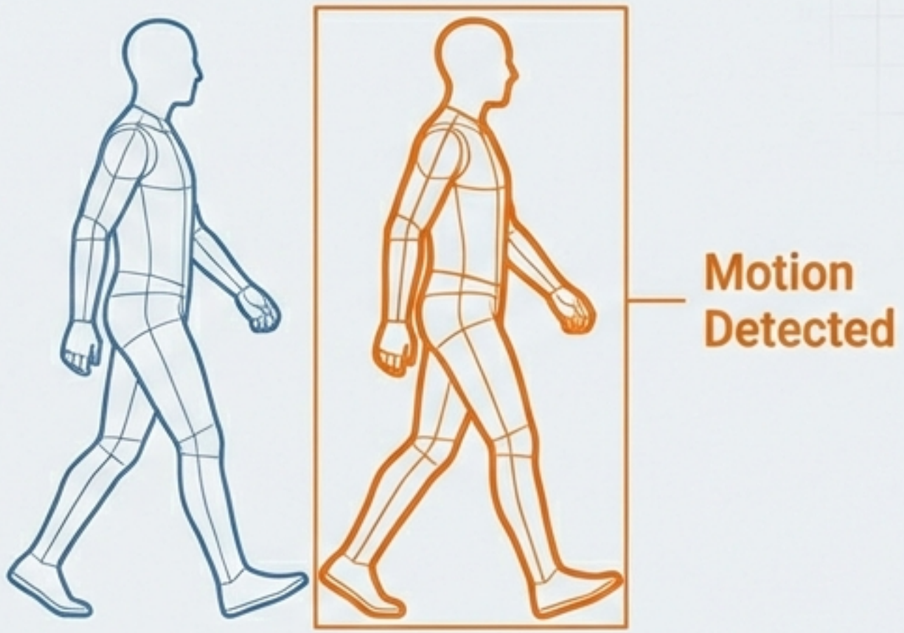
The Pipeline Step 2: Alignment & De-Ghosting

The Challenge



Between burst frames, hands shake and subjects move. Merging without alignment creates “Ghosting” artifacts.

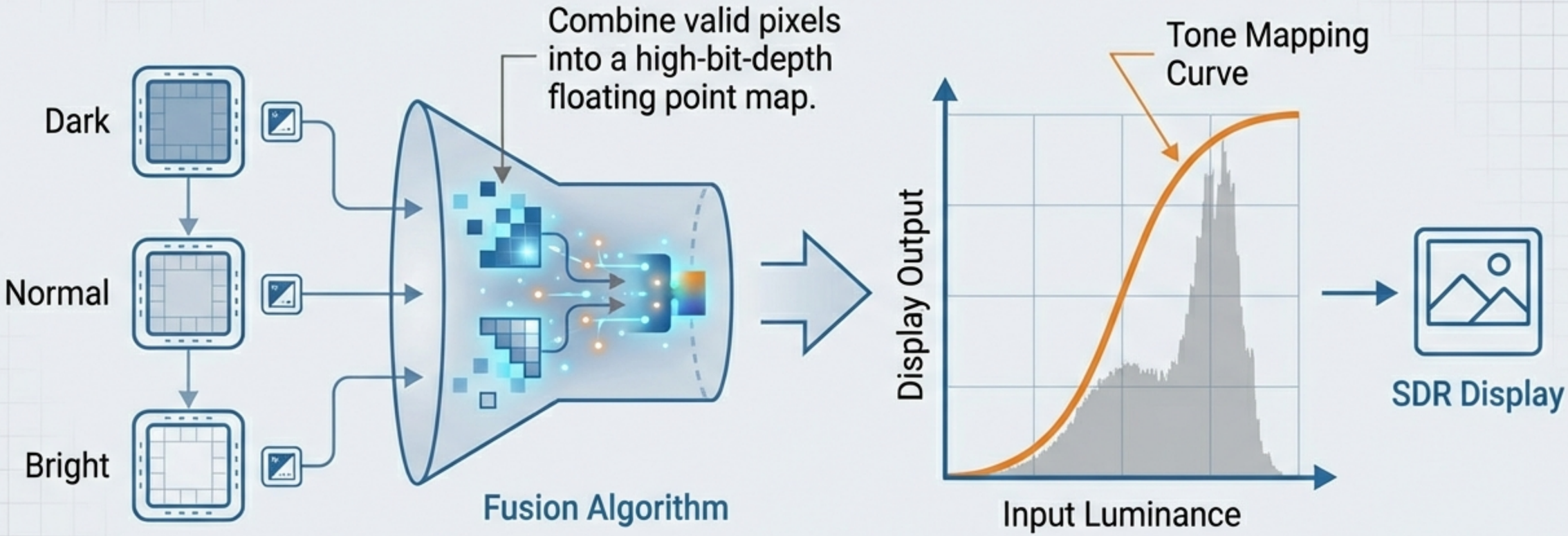
The Solution



Algorithms align the background (Global Alignment) and detect moving pixels (Motion Segmentation). The system selects a “reference frame” for the moving subject to prevent transparency.

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The Pipeline Step 3: Fusion & Tone Mapping



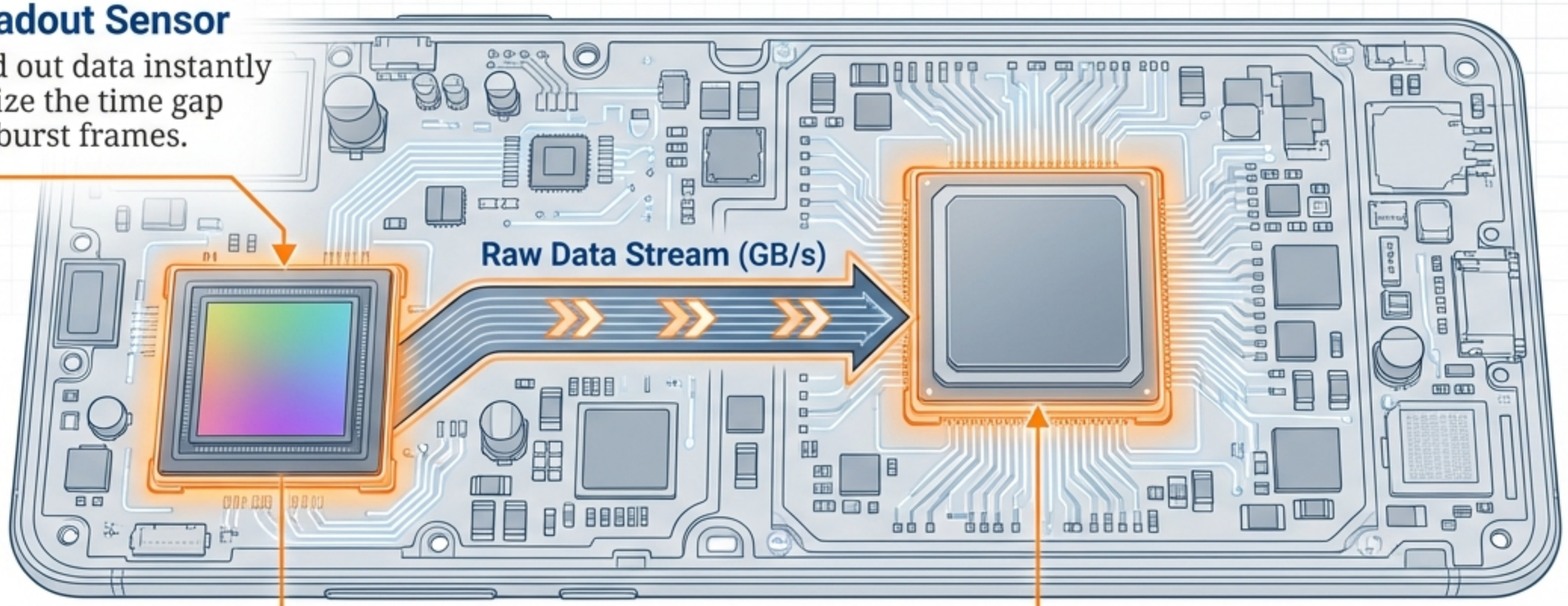
Fusion creates a file with more dynamic range than a screen can show. Tone Mapping compresses this range back down to Standard Dynamic Range (SDR) while mathematically boosting local micro-contrast to keep the image looking punchy.

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Hardware Enablers: Speed & Intelligence

Fast Readout Sensor

Must read out data instantly to minimize the time gap between burst frames.



Fast Readout Sensor

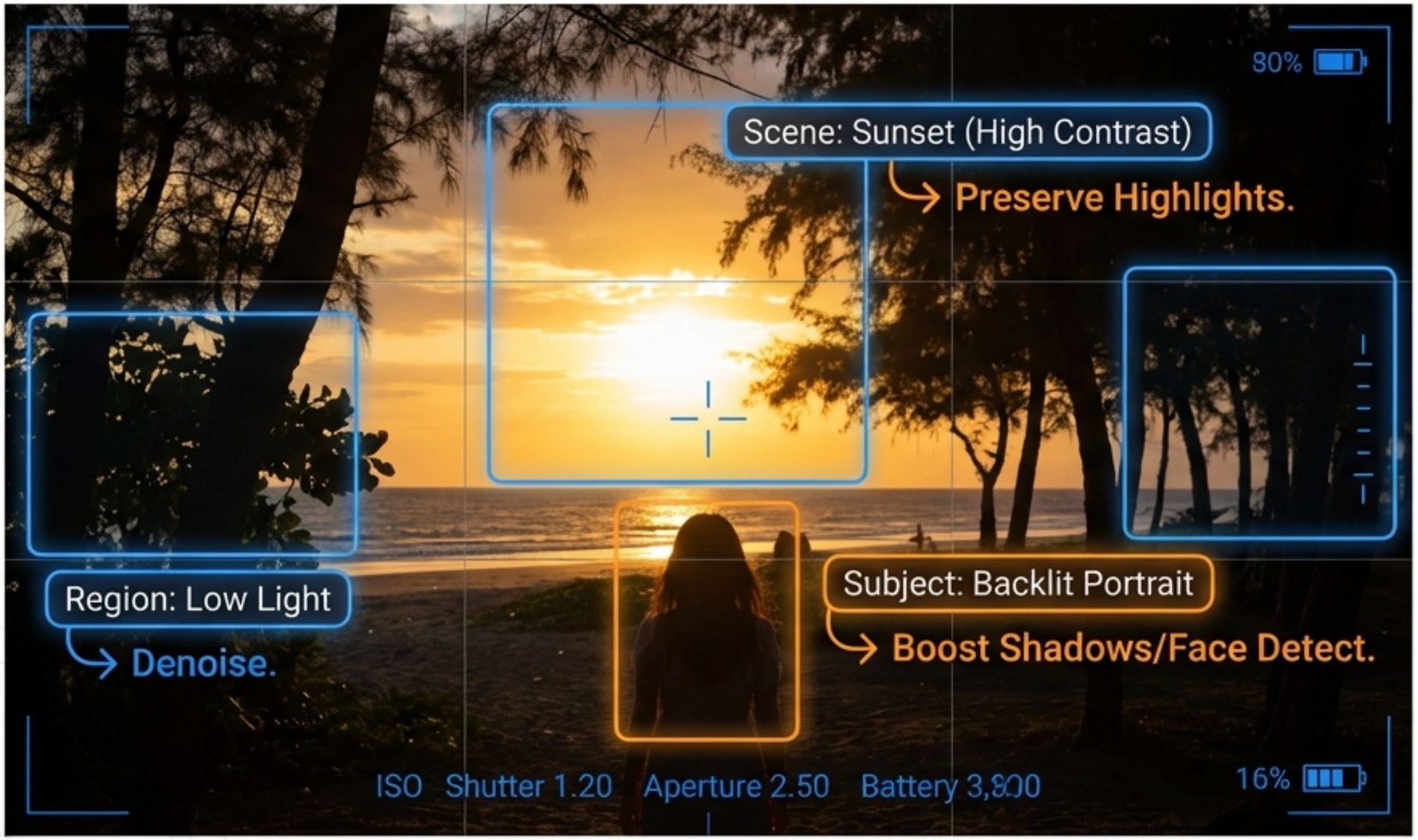
Must read out data instantly to minimize the time gap between burst frames.

ISP + NPU (Neural Processing Unit)

Dedicated silicon for matrix operations. Unlike a CPU, the NPU handles the massive computational load of merging 4K frames in milliseconds.

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Software Intelligence: Scene Recognition



Contextual Awareness

The camera isn't just capturing light; it's understanding the scene.

Deep learning models analyze the live preview to trigger HDR automatically when high contrast is detected.

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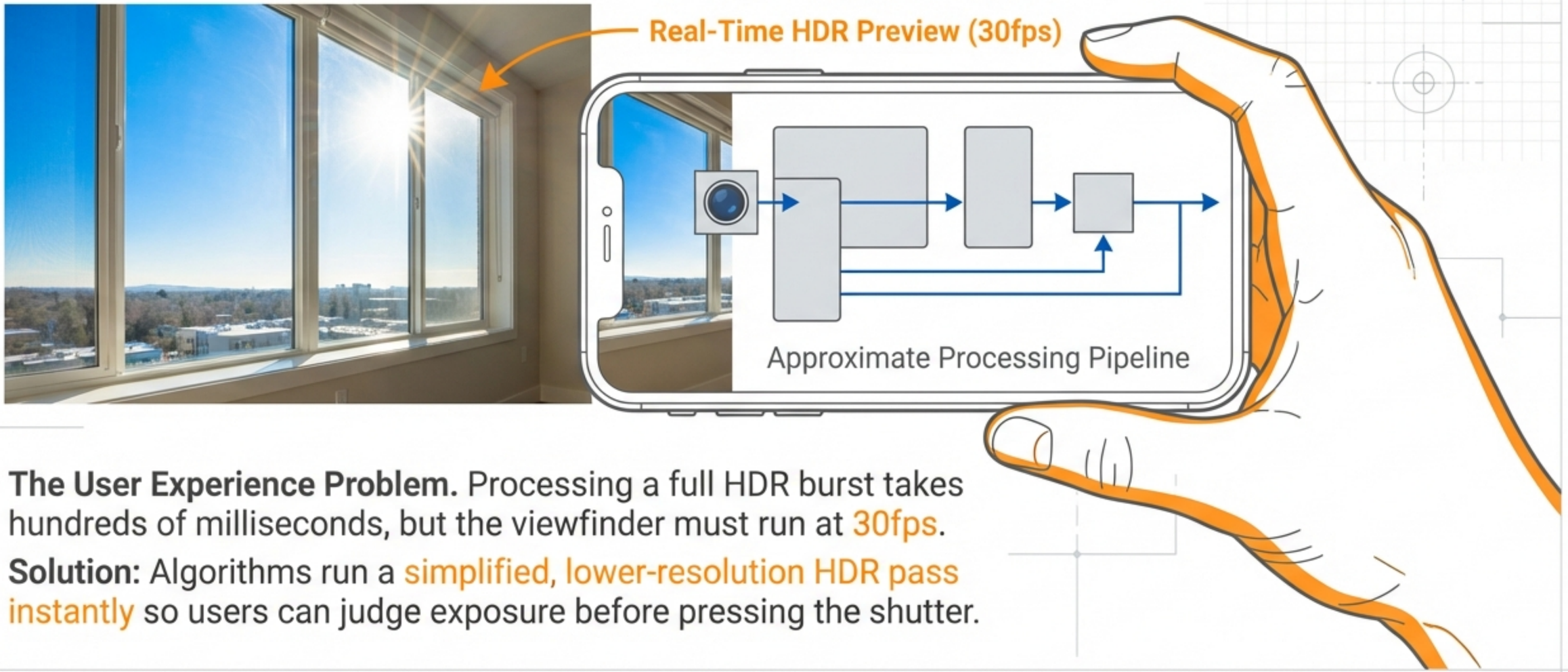
Deep Learning Augmentation



- **Denoising:** Identifying and removing digital noise in shadows.
- **Restoration:** Reconstructing color fidelity in clipped areas.
- **Detail Enhancement:** Hallucinating plausible textures based on learned patterns.

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The WYSIWYG Challenge

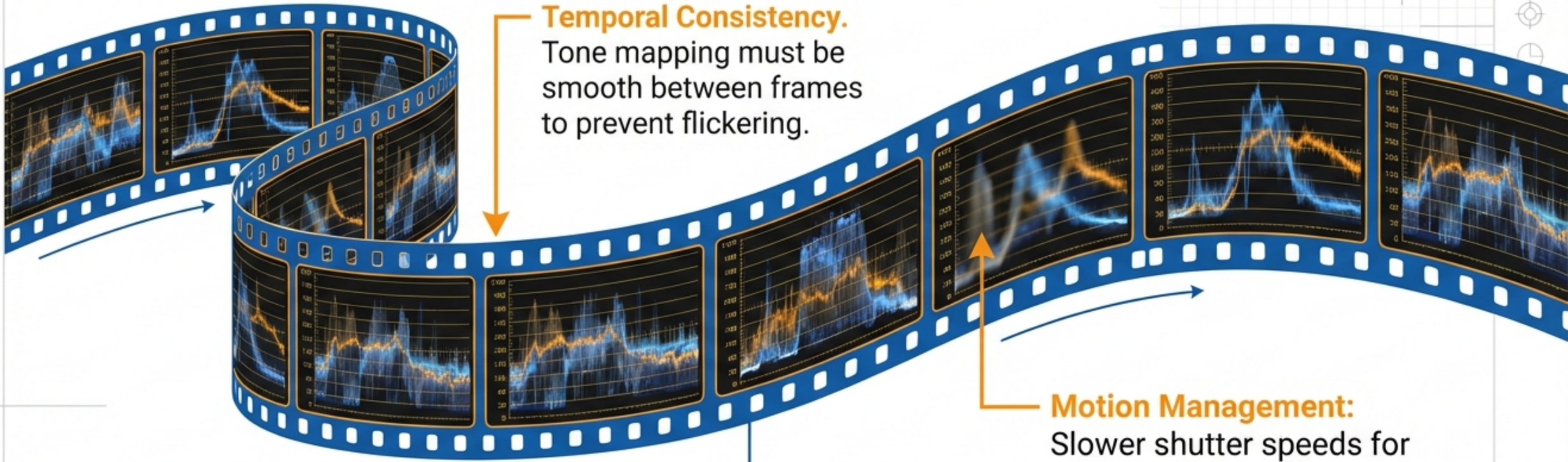


The User Experience Problem. Processing a full HDR burst takes hundreds of milliseconds, but the viewfinder must run at 30fps.

Solution: Algorithms run a simplified, lower-resolution HDR pass instantly so users can judge exposure before pressing the shutter.

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HDR in Video: The Exponential Challenge



Temporal Consistency.
Tone mapping must be smooth between frames to prevent flickering.

Motion Management:
Slower shutter speeds for shadow recovery can introduce unwanted motion blur.

Continuous Processing @ 60 Frames Per Second

Video requires applying the full HDR pipeline to every single frame instantly, generating massive heat and battery load.

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Standards & Ecosystem

Display Pipelines

8-bit SDR



10-bit HDR



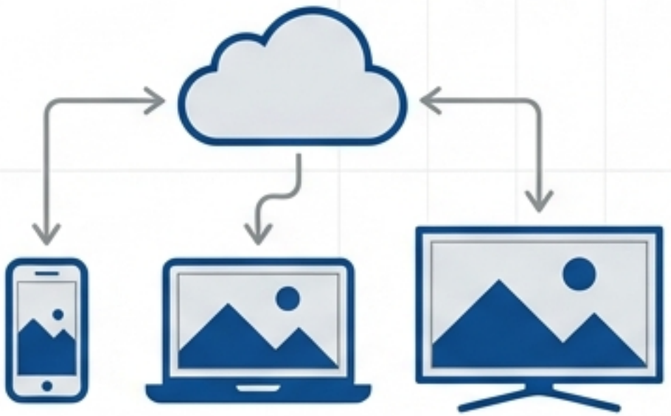
Billions of Colors vs **Millions**

Formats



Dynamic Metadata tells the screen how to tune brightness frame-by-frame.

Cross-Device



Cloud optimization ensures the image looks correct on any display.

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Core Challenges Remaining



Motion Artifacts

Fast-moving objects still challenge de-ghosting algorithms, leading to blurred or double edges.



Thermal Throttling

Heavy computational load generates heat, limiting how long HDR video can be recorded.



Power Consumption

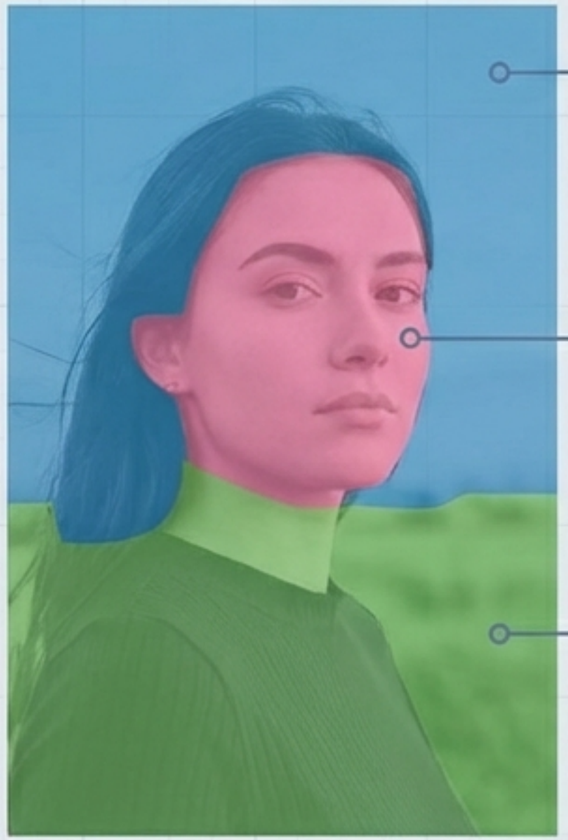
Continuous multi-frame processing is a significant drain on battery life.

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Future Directions: Semantic & Generative

Digital Twin



Sky Contrast — High

Face Softness — High

Grass Detail — Low

Semantic-Aware Tuning

Applying different tone mapping curves to different objects. Sky gets high contrast; Face gets soft contrast.

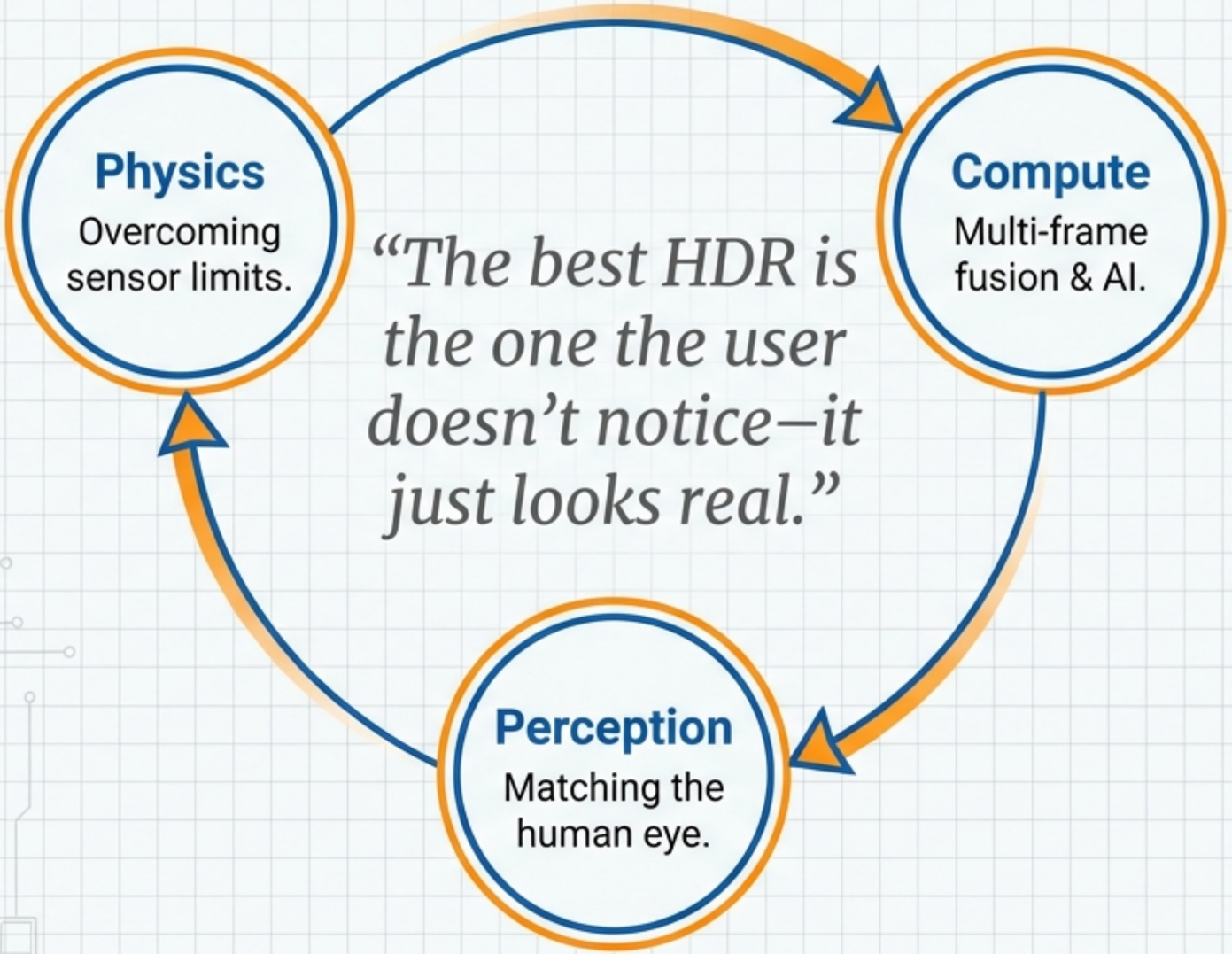


Generative AI: Using AI to ‘imagine’ and fill in highlight details that were physically lost by the sensor.

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Summary: The Invisible Technology



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